

**P08 – AI-Powered Research Assistant**

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|--------------------------|-------------------------|
| <b>Group Formal Tag:</b> | SOFT3888_TU08_01        |
| <b>Tutor Name:</b>       | Michael de Francesco    |
| <b>Client Name:</b>      | Vecton AI (Bocheng Han) |

**Group members details**

| <b>No.</b> | <b>Student Name</b>    | <b>Student ID</b> | <b>Student Unikey</b> | <b>XP-Roles</b>                           |
|------------|------------------------|-------------------|-----------------------|-------------------------------------------|
| <b>1</b>   | Joshua Morton          | 530500735         | jmor4801              | Programmer<br>Tracker<br>Customer Liaison |
| <b>2</b>   | Prithish Lahiri        | 530609175         | plah4259              | Programmer<br>Manager<br>Customer Liaison |
| <b>3</b>   | Fiona Lee              | 540880966         | flee0409              | Programmer<br>Tracker                     |
| <b>4</b>   | Kang Le<br>Charles Lim | 540375026         | klim0039              | Programmer<br>Doomsayer                   |
| <b>5</b>   | Felicia Chen           | 530532079         | lche6131              | Programmer<br>Manager                     |
| <b>6</b>   | Xiyuan Pan             | 530667566         | xpan0988              | Programmer<br>Tester                      |
| <b>7</b>   | Xingyu Liu             | 510015523         | xliu3209              | Programmer<br>Tester                      |

Attendances: *Joshua Morton, Prithish Lahiri, Fiona Lee, Kang Le Charles Lim, Felicia Chen, Xiyuan Pan, Xingyu Liu*

Apologies: *none*

Submission Date: 30/08/26

**Weekly Group Meeting Minutes**

Time: *1st Meeting of week 4 (Post presentation in tutorial), 25th August*

Venue: *Tutorial room (J02.03.315 PNR Building Learning Studio 315)*

Meeting Minute Taker: *Prithish Lahiri*

**Attendances:** *Joshua Morton, Prithish Lahiri, Fiona Lee, Kang Le Charles Lim, Felicia Chen, Xiyuan Pan, Xingyu Liu*

**Apologies:** *none*

| # | Agenda Item              | Description/<br>Comments                                                                                                                                                                               | Decision/Action                                     | Who?        | Items for escalation     |
|---|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------|--------------------------|
| 1 | What has been completed? | Completed and presented in class, the project proposal report and presentation covered the project's overview, the research that was done, and the future plan.                                        | Maintain as the basis for further work.             | All members | No immediate escalation. |
| 2 | What is in progress?     | To create the database for the AI research assistant, each team member is gathering information on their own. Members are investigating various LLM models concurrently to choose one for the project. | Maintain parallel data collection and LLM research. | All members | No immediate escalation. |
| 3 | What is working well?    | Members of the team communicate well with one another, sharing findings and cooperating with one another.                                                                                              | Maintain your present methods of communication.     | All members | No immediate escalation. |

|   |                                                     |                                                                                                                       |                                                                |             |                          |
|---|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-------------|--------------------------|
| 4 | What needs improvement?                             | There is currently no conflict because the team is concentrating on creating a functional MVP as quickly as possible. | Keep your current focus and pace.                              | All members | No immediate escalation. |
| 5 | Reminders                                           | Minutes of meetings must be kept current. The weekly client meeting needs to be scheduled.                            | Set up a meeting with the client for this week.                | All members | No immediate escalation. |
| 6 | What needs to be completed before the next meeting? | At the next client meeting, prepare a list of questions to ask the client.                                            | Draft enquiries for clients                                    | All members | No immediate escalation. |
| 7 | <b>What's Next</b>                                  | Continue looking into LLM options and start organising the strategy for gathering data.                               | Proceed with the planning of data collection and LLM research. | All members | No immediate escalation. |

#### **Main Discussions:**

- The group went over the finished project proposal report and presentation that was given in class.
- In order to divide the workload of comparing options given the client's no-budget constraint, the team decided that each member would independently choose and investigate an LLM model.
- The group talked about how to start creating the researcher/publication database, including dividing member profiles and using a common method of collection.
- The group talked about creating a list of enquiries to present at the next client meeting.

#### **Key Takeaway:**

- Database construction and LLM research can be divided among team members and carried out concurrently.
- .Assigning a models to each individual expedites the final comparison and prevents redundant research.
- To obtain clearer guidance, the team should bring specific questions to the client meeting instead of open-ended ones.

**What Happened:**

- The Project Proposal Report and Presentation were finished and presented in the tutorial last week.
- Database development and LLM research are now carried over as ongoing tasks that are divided among team members.

**What's Next:**

- Joshua Morton is in charge of investigating Llama and Copilot as potential LLM choices.
- Claude and Gemini are being investigated as potential LLM options by Prithish Lahiri.
- Fiona Lee is in charge of investigating Mistral as a potential LLM choice.
- Felicia Chen is in charge of investigating Voyage AI and Cohere as potential LLM choices.
- Xiyuan Pan is in charge of investigating Kimi and Qwen as potential LLM choices.
- Each person is in charge of starting their own data collection for the shared database.
- It is everyone's responsibility to prepare questions for the next client meeting.

## Weekly Client Meeting Minutes

Time: 28th August 9:00pm

Venue: Google Meet

Meeting Minute Taker: Prithish Lahiri

**Attendances:** *Joshua Morton, Prithish Lahiri, Fiona Lee, Kang Le Charles Lim, Felicia Chen, Xiyuan Pan, Xingyu Liu*

**Apologies:** *none*

| # | Agenda Item              | Description/ Comments                                                                                                      | Decision/Action                                                                                         | Who?        | Items for escalation     |
|---|--------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------|--------------------------|
| 1 | What has been completed? | The current project plan, which included the suggested tech stack and data collection progress, was presented by the team. | The plan as presented was accepted by the client.                                                       | All members | No immediate escalation. |
| 2 | What is in progress?     | Prior to this meeting, LLM research was being conducted concurrently with data collection.                                 | The client suggested that for the time being, data collection should take precedence over LLM research. | All members | No immediate escalation. |
| 3 | What is working well?    | The client showed no opposition or pushback to the team's plan or progress.                                                | Maintain your current pace and strategy.                                                                | All members | No immediate escalation. |
| 4 | What needs improvement?  | The client did not bring up any concerns during this meeting.                                                              | N/A                                                                                                     | All members | No immediate escalation. |

|   |                                                     |                                                                                                    |                                                                   |             |                          |
|---|-----------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------|--------------------------|
| 5 | Reminders                                           | From now on, there will be regular weekly client meetings.                                         | Every Thursday at 4:30 p.m. is the confirmed weekly meeting time. | All members | No immediate escalation. |
| 6 | What needs to be completed before the next meeting? | As instructed by the client, keep gathering data as the top priority.                              | Prioritise data collection over other LLM research                | All members | No immediate escalation. |
| 7 | <b>What's Next</b>                                  | Continue gathering data; once data collection has advanced sufficiently, start LLM research again. | Priority should be data collection                                | All members | No immediate escalation. |

#### **Main Discussions:**

- The team informed the client about data collection and LLM research advancements while presenting the current tech stack.
- After reviewing the plan, the client had no issues or requests for revisions.
- The client suggested that the team put database development and data collection ahead of further LLM research for the time being.
- Going forward, the team and client agreed to a set weekly meeting schedule.

#### **Key Takeaways**

- The client is happy with the project's current direction and tech stack.
- As directed explicitly by the client, data collection and database construction should take precedence over LLM research.
- The new schedule for weekly client meetings is Thursdays at 4:30 p.m.

#### **What Happened:**

- The team presented its tech stack and progress to the client last week, and they were approved without any changes being asked for.
- The client deprioritized LLM research until later and shifted the team's attention to database construction and data collection as the top priorities.

**Whats Next:**

- Until further notice, everyone must prioritise database construction and data collection over LLM research.
- Everyone must show up for the recently confirmed weekly client meeting, which takes place every Thursday at 4:30 p.m.

**Out of Tutorial meeting**

Time: 3rd Meeting of week 4 (Out of tutorial group meeting), 29th August, 8:00pm

Venue: Instagram Call (25mins)

Meeting Minute Taker: Prithish Lahiri

**Attendances:** Joshua Morton, Prithish Lahiri, Fiona Lee, Kang Le Charles Lim, Felicia Chen, Xiyuan Pan, Xingyu Liu

**Apologies:** none

| # | Agenda Item              | Description/<br>Comments                                                                                                                                                                                                                                                                                     | Decision/Action                                                                                         | Who?        | Items for<br>escalation                                                         |
|---|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------------------------------------------|
| 1 | What has been completed? | Finalised the tech stack and the data structure was clarified. The data collection tasks were assigned to each team member. At the same time, the core MVP research and development began, resulting in a draft of the search and ranking model design, and evaluation methods for each layer were obtained. | Begin database building and search/ranking model development                                            | All members | No immediate escalation.                                                        |
| 2 | What is in progress?     | A shared Google Sheet is being used to collect data, with 150 profiles divided roughly 20–25 per person across the nine universities, with a focus on CS/IT. A feasibility comparison between RAG and GraphRAG is underway.                                                                                  | As instructed by the client, keep gathering data as a top priority and keep comparing RAG and GraphRAG. | All members | Before deciding on a retrieval strategy, GraphRAG's viability must be verified. |

|   |                                                     |                                                                                                                                                                                                                                             |                                                                                        |             |                          |
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| 3 | What is working well?                               | Each member is given a specific technical deliverable (data source investigation, frontend structure, database schema, backend architecture, search/ranking design, RAG/GraphRAG comparison), and there is clear individual task ownership. | For the upcoming sprint, maintain the current task allocation structure.               | All members | No immediate escalation. |
| 4 | What needs improvement?                             | n/a                                                                                                                                                                                                                                         | n/a                                                                                    | All members | No immediate escalation. |
| 5 | Reminders                                           | CS/IT coverage should be given top priority when gathering data, and all nine universities should be included. Each entry must include the source URL and the collection date.                                                              | Observe the established provenance-tracking guidelines and data collection field list. | All members | No immediate escalation. |
| 6 | What needs to be completed before the next meeting? | Complete individual LLM research, keep gathering data to reach the 150-profile goal, and complete the database schema and search/ranking proposal for team approval.                                                                        | Prior to the next meeting, assign each person a completion goal.                       | All members | No immediate escalation. |
| 7 | <b>What's Next</b>                                  | Once the initial data collection is complete, move on to data cleaning and validation planning; complete the RAG                                                                                                                            | Start organising the validation and cleaning stage.                                    | All members | No immediate escalation. |

|  |  |                              |  |  |  |
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|  |  | vs. GraphRAG recommendation. |  |  |  |
|--|--|------------------------------|--|--|--|

### Main Discussions:

- Completed the data collection plan, which prioritises CS/IT and uses Google Sheet storage to store 150 profiles, divided roughly 20–25 per person across 9 universities.
- Verified data fields and sourcing order (OpenAlex/ORCID/Crossref after university sites).
- Examined the search/ranking design and the draft database schema.

### Key Takeaways

- The plan for gathering data has been completed and is prepared for implementation.
- The search/ranking design and database schema are prepared for deployment.
- After sufficient data is gathered, cleaning and validation planning takes place.

### What Happened:

- Building databases and gathering data continued to be the client's top priorities.
- The team reviewed the schema/ranking designs and completed the concrete collection plan.

### Whats Next:

- Everyone: gather the designated 20–25 profiles in order to reach the 150-profile goal.
- Fiona Lee: Complete the suggested database schema.
- Felicia Chen: Finalise the search/ranking design proposal
- Xiyuan Pan: Complete the recommendation between RAG and GraphRAG.
- Prithish: finalise front end structure, make full wireframe and present to group for approval

## Individual Progress Report

Name: Joshua Morton

Student ID: 530500735

### Goals

- To have better clarification from the client on which tech stack tools to use
- To have a clear draft of what the database will look like
- Look into Llama and Copilot as LLM tools

### Progress Made *[What progress was made? Are the goals satisfied?]*

- During the group meeting with the client he showed approval of all our tech stack tool choices
- A draft ER Diagram was written up for the database
- The feasibility of using each LLM tool was looked into

**Blockers to Progress/Issues** *[What stopped you from making progress? What issues need group or client decisions?]*

- Despite showing approval of our tech stack our client did not provide much guidance
- Hard to organise time to meet with group
- Lots of work outside of the project

Name: Prithish Lahiri

Student ID: 530609175

**Goals** *[What were your goals? These should be the ones outlined in last week's minutes]*

- Compare llm models for future implementation
- Planning data collection methods
- Building frontend structure, identify userflows and create wireframes

**Progress Made** *[What progress was made? Are the goals satisfied?]*

- Identified key userflows and made a skeleton of front end structure, yet to finish wireframe
- Data collection process and academic profile search finalised

**Blockers to Progress/Issues** *[What stopped you from making progress? What issues need group or client decisions?]*

- Multiple commitments including thesis work
- Tough to find common times amongst all members

Name: Fiona Lee

Student ID: 540880966

**Goals** *[What were your goals? These should be the ones outlined in last week's minutes]*

- Ensure everyone is on track with the current progress
- Start planning database schema and data collection
- Research and compare LLM choice

**Progress Made** *[What progress was made? Are the goals satisfied?]*

- Planned data collection method, initial storage using google spreadsheet
- Researched the pros and cons of several LLMs
- Finished the scope statement and project proposal presentation

**Blockers to Progress/Issues** *[What stopped you from making progress? What issues need group or client decisions?]*

*Communication and common time to meet is often tough*

Name: Felicia Chen

Student ID: 530532079

**Goals** *[What were your goals? These should be the ones outlined in last week's minutes]*

Research and evaluate potential approaches for semantic search, keyword search, and paper ranking, aligned with project requirements (Functional Requirements 3.3 & 3.4).

**Progress Made** *[What progress was made? Are the goals satisfied?]*

Draft a preliminary search & ranking design proposal for team discussion, covering:

1. Retrieval strategy (hybrid keyword + semantic)
2. Re-ranking logic (academic impact, recency, explainability)
3. Evaluation metrics

**Blockers to Progress/Issues** *[What stopped you from making progress? What issues need group or client decisions?]*

No major blockers to my individual progress. However, several team-level decisions are needed to move to implementation:

1. Embedding model selection: Decision on which sentence-transformer model to use (e.g. all-MiniLM-L6-v2 for speed vs. BGE-large for quality).
2. Venue weighting: Whether to include venue ranking (CCF/JCR) as a factor — this affects data collection effort.

Name: Kang Le Charles Lim

Student ID: 540375026

**Goals** *[What were your goals? These should be the ones outlined in last week's minutes]*

Goals this week are to finish the presentation and to finalise project scope and requirements.

**Progress Made** *[What progress was made? Are the goals satisfied?]*

Met the client this week to confirm meeting times and show project's progress. The group has finalized the project's direction and we have now begun collecting data.

**Blockers to Progress/Issues** *[What stopped you from making progress? What issues need group or client decisions?]*

We still have not decided on the metrics used to rank the research. We have not decided on what llm we should use yet.

Name: Xiyuan Pan

Student ID: 530667566

**Goals**

- Finalise the initial technical direction for the GraphRAG component.
- Coordinate with the client and clarify remaining project requirements.
- Review the proposed academic data model and determine how it can support GraphRAG retrieval.
- Prepare for independent development of the GraphRAG module.

### **Progress Made**

- Contacted the client and coordinated the time for the Week 4 client meeting.
- Reviewed the team's proposed data-collection approach and database schema, including researcher, publication, institution, topic, source and evidence entities.
- Confirmed a metadata/schema-based approach for academic publication data rather than relying on full-text storage as a core requirement.
- Developed an initial GraphRAG design based on relationships such as authorship, affiliation, citation and research topics.
- Identified how the GraphRAG component can be developed independently using mock data and later integrated with the Supabase/PostgreSQL database through a defined interface.
- Identified remaining design decisions around ranking signals, including semantic relevance, citation influence and recency.

### **Blockers to Progress/Issues**

- The final weighting and strategy for ranking researchers and publications still requires confirmation or team evaluation.
- The specific free-tier LLM/embedding provider has not yet been selected because the project has no allocated paid API budget.
- Full GraphRAG integration will depend on the final database implementation and availability of cleaned project data; however, the retrieval module can be developed independently using mock data in the meantime.

Name: Xingyu Liu

Student ID:

**Goals** *[What were your goals? These should be the ones outlined in last week's minutes].*

Define the initial researcher profile structure required for the Australian Researcher Directory.

Identify the key information fields needed to describe each researcher profile for the shared database.

Produce an initial researcher profile data schema as the team's deliverable.

**Progress Made** *[What progress was made? Are the goals satisfied?]*

Drafted an initial researcher profile schema outlining the core fields needed to describe each academic (e.g. name, affiliation, department, research area/topics, publication list, contact/profile URL, source URL and date collected).

Aligned the proposed schema with the team's provenance-tracking requirements, ensuring each entry can trace back to its original source (university page, OpenAlex/ORCID/Crossref).

Cross-checked the schema against the group's data collection plan (9 universities, CS/IT focus, ~150 profiles) to confirm it can support consistent data entry across all team members.

**Blockers to Progress/Issues** [*What stopped you from making progress? What issues need group or client decisions?*]

Final field list still needs team review/approval to ensure consistency once data collection scales up.

Some universities present researcher information differently (varying formats/available fields), which may require the schema to accommodate optional or university-specific fields.